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A	 PT. ABB Sakti Industri Jalan Gajah Tunggal km 1.0 Tangerang 15135, Indonesia								A																																												
B									B																																												
C	Project Name : MV Switchgear for MNA Serang Plant Switchgear Name : MVD-27 20kV Tank Farm Panel Type : UNISEC Customer Name : PT Multimas Nabati Asahan Contract No. : BE318042 TITLE : SLD & GA Drawing								C																																												
D									D																																												
E									E																																												
F	<table><tr><td></td><td></td><td></td><td></td><td>Date Prep.</td><td>15.11.2018</td><td>PROJECT TITLE</td><td rowspan="2">ABB EPMV</td><td rowspan="2">COVER SHEET SLD & GA Drawing MVD-27 20kV Tank Farm</td><td>Scale</td><td></td><td>= H00</td></tr><tr><td>1</td><td>Issued for Approval</td><td>18.12.2018</td><td>YR</td><td>Prepared</td><td>YR</td><td>MV Switchgear for MNA Serang Plant</td><td>Customer</td><td>PT Multimas Nabati Asahan</td><td>& GAD</td></tr><tr><td>0</td><td>First Issued</td><td>19.11.2018</td><td>YR</td><td>Checked</td><td>AW</td><td></td><td rowspan="2">PT. ABB Sakti Industri</td><td></td><td>Drawing Nbr.</td><td></td><td>Sh. 1</td></tr><tr><td>Ind.</td><td>Revision</td><td>Date</td><td>Name</td><td>Approved</td><td>AI</td><td>Based on: BE318042</td><td>Customer Doc. Nbr.</td><td></td><td>Cont. 16</td></tr></table>												Date Prep.	15.11.2018	PROJECT TITLE	ABB EPMV	COVER SHEET SLD & GA Drawing MVD-27 20kV Tank Farm	Scale		= H00	1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant	Customer	PT Multimas Nabati Asahan	& GAD	0	First Issued	19.11.2018	YR	Checked	AW		PT. ABB Sakti Industri		Drawing Nbr.		Sh. 1	Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.		Cont. 16	F
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A	SHEET INDEX								A				
B	PAGE		PAGE NAME			REVISION							
	==MVD-27=H00&GAD/1		COVER SHEET			0	1						
	==MVD-27=H00&GAD/2		INDEX OF SHEETS			0	1						
	==MVD-27=H00&GAD/3		GENERAL CHARACTERISTICS			0	1						
	==MVD-27=H00&GAD/4		REFERENCE DESIGNATIONS			0	1						
	==MVD-27=H00&GAD/5		REFERENCE DESIGNATIONS			0	1						
	==MVD-27=H00&GAD/6		REFERENCE DESIGNATIONS			0	1						
	==MVD-27=H00&GAD/7		REFERENCE DESIGNATIONS			0	1						
	==MVD-27=H00&GAD/8		SINGLE LINE DIAGRAM			0	1						
	==MVD-27=H00&GAD/9		INSTALLATION RULES			0	1						
	==MVD-27=H00&GAD/10		FRONT VIEW & GA			0	1						
	==MVD-27=H00&GAD/11		FOUNDATION FRAMES			0	1						
	==MVD-27=H00&GAD/12		FOUNDATION FRAMES DETAILS			0	1						
	==MVD-27=H00&GAD/13		FOUNDATION FRAMES DETAILS			0	1						
	==MVD-27=H00&GAD/14		SECTION VIEW			0	1						
	C	==MVD-27=H00&GAD/15		SECTION VIEW			0	1					
==MVD-27=H00&GAD/16		SECTION VIEW			0	1							
D									D				
E									E				
F									F				
				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	INDEX OF SHEETS SLD & GA Drawing MVD-27 20kV Tank Farm	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW			PT. ABB Sakti Industri	Drawing Nbr.			Sh. 2	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.				Cont. 16		
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A	CHARACTERISTICS (IN COMPLIANCE WITH STANDARD IEC 62271-200)				NOTES				A
B	SWITCHGEAR VERSION			= UNISEC	COLOUR OF FRONT DOORS = RAL 7035				B
	RATED VOLTAGE (Un)			= 24 kV	SIDEWALLS PAINTED = YES				
	OPERATING VOLTAGE			= 20 kV	SIDES = L+R				
	RATED FREQUENCY (fr)			= 50 Hz					
	RATED LIGHTNING IMPULSE WITHSTAND VOLTAGE (Up)			= 125 kV					
C	RATED POWER FREQUENCY WITHSTAND VOLTAGE (Ud)			= 50 kV					C
	RATED CURRENT OF MAIN BUSBARS (Ir)			= 630 A					
	RATED SHORT-TIME WITHSTAND CURRENT (Ik)			= 20 kA					
	RATED PEAK WITHSTAND CURRENT (Ip)			= 50 kA					
	RATED DURATION OF SHORT CIRCUIT (tk)			= 1 s					
D	INTERNAL ARC CLASSIFICATION (IAC)			= AFLR / 16kAx1s					D
	AMBIENT CONDITION			= NORMAL					
	AMBIENT AIR TEMPERATURE			= -5°C...+40°C					
	DEGREE OF PROTECTION (OPERATION SEATS EXCLUDED)			= IP3X					
	DEGREE OF PROTECTION WITH OPEN DOORS			= IP2X					
E	RATED SUPPLY VOLTAGE OF CONTROL AND SIGNALLING CIRCUITS (Ua)			= 230VAC					E
	RATED SUPPLY VOLTAGE OF SPRING CHARGING MOTOR (Ua)			= 230VAC					
	RATED SUPPLY VOLTAGE OF LIGHTING AND HEATING CIRCUITS (Ua)			= 230VAC					
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS			= 450/750V					
	TYPE OF CABLE FOR LOW VOLTAGE CONDUCTORS			= PVC					
F	CROSS-SECTION OF CONDUCTORS FOR CURRENT CIRCUITS			= 2.5 mm²/BLACK					F
	CROSS-SECTION OF CONDUCTORS FOR VOLTAGE CIRCUITS			= 1.5 mm²/BLACK					
	CROSS-SECTION OF CONDUCTORS FOR MOTORS			= 2.5 mm²/BLACK					
	CROSS-SECTION OF OTHER CONDUCTORS (CONTROL AND SIGNALLING CIRCUITS)			= 1.5 mm²/BLACK					
	CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS			= 1.5 mm²/BLACK					
CROSS-SECTION OF CONDUCTORS FOR INTERCONNECTIONS OF SUPPLY VOLTAGE			= 2.5 mm²/BLACK						
COMMUNICATION CABLE			= N/A						
1 Issued for Approval 18.12.2018 YR Date Prep. 15.11.2018 PROJECT TITLE MV Switchgear for MNA Serang Plant Based on: BE318042 ABB EPMV PT. ABB Sakti Industri GENERAL CHARACTERISTICS SLD & GA Drawing MVD-27 20kV Tank Farm Scale Customer Drawing Nbr. Customer Doc. Nbr. = H00 PT Multimas Nabati Asahan Sh. 3 Cont. 16									
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FOR IDENTIFICATION OF CONDUCTORS THE "LOCAL END CONNECTION LABELLING" SYSTEM IS USED, IN COMPLIANCE WITH STANDARD IEC 62491 PARAGRAPH 6.2. THIS SYSTEM FORESEES THAT THE MARKING OF A CONDUCTOR END SHOWS THE DESIGNATION OF THE TERMINAL TO WHICH IT IS CONNECTED. IN COMPLIANCE WITH STANDARD IEC 61666 EACH CONDUCTOR IS IDENTIFIED WITH THE REFERENCE DESIGNATION OF THE CONNECTED ELECTRICAL COMPONENT FOLLOWED BY SIGN ":" (COLON), FOLLOWED BY THE TERMINAL DESIGNATION. FOR EXAMPLE THE CONDUCTOR CONNECTED TO TERMINAL "10" OF THE COMPONENT WITH DESIGNATION "-XA" HAS THE MARKING "-XA:10". THE CONDUCTOR MARKING IS PLACED IN SUCH A WAY TO READ IT ALWAYS FROM THE TERMINAL TOWARDS THE CONDUCTOR, AS SHOWN BELOW

-XDA:10

91011

-XDA:10

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A	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS								A
	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)								
	DESIGNATION	DESCRIPTION							
B	-AA	MULTIFUNCTION UNIT (CENTRAL UNIT)			-BGB5	POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE ENERGIZED			
	-BAD1,-BAD4	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1			-BGB6	POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE EXCLUDED MECHANICALLY			
	-BAD2,-BAD5	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2			-BGB7	POSITION SWITCHES OF CIRCUIT-BREAKER ON SWITCHGEAR			
	-BAD3,-BAD6	CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3			-BGB8...-BGB10	POSITION SWITCHES OF CIRCUIT-BREAKER			
	-BAR	VOLTAGE PROTECTION RELAY			-BGB11	POSITION SWITCH OF CIRCUIT-BREAKER OPERATING FOR MANUAL OPENING			
	-BAS1	VOLTAGE SENSOR LOCATED ON PHASE L1			-BGB12	PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN OPEN POSITION			
	-BAS2	VOLTAGE SENSOR LOCATED ON PHASE L2			-BGB13	PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN CLOSED POSITION			
	-BAS3	VOLTAGE SENSOR LOCATED ON PHASE L3			-BGD	POSITION SWITCH OF L.V. INSTRUMENT COMPARTMENT DOOR.			
	-BAT1	VOLTAGE TRANSFORMER LOCATED ON PHASE L1			-BGD1	POSITION SWITCH OF CIRCUIT-BREAKER COMPARTMENT DOOR			
	-BAT2	VOLTAGE TRANSFORMER LOCATED ON PHASE L2			-BGD2	POSITION SWITCH OF CABLE COMPARTMENT DOOR			
C	-BAT3	VOLTAGE TRANSFORMER LOCATED ON PHASE L3			-BGE1	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION			B
	-BAT4	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1			-BGE2	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN CLOSED POSITION			
	-BAT5	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2			-BGE3	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)			
	-BAT6	VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3			-BGE4	POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE ACTUATED BY ELECTRIC MOTOR			
	-BCD	DIFFERENTIAL PROTECTIVE RELAY			-BGE5	POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 ACTUATED BY ELECTRIC MOTOR			
	-BCF	FEEDER PROTECTION RELAY			-BGE7	POSITION SWITCH SIGNALLING EARTHING SWITCH NOT IN MANUAL OPERATION			
	-BCG	GENERATOR PROTECTION RELAY			-BGE8	POSITION SWITCHES OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH- DISCONNECTOR -QBS			
	-BCM	MOTOR PROTECTION RELAY			-BGE11	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION			C
	-BCN	NEUTRAL (RESIDUAL) CURRENT TRANSFORMER			-BGE12	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN CLOSED POSITION			
	-BCP	TRANSFORMER PROTECTION RELAY			-BGE13	POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)			
D	-BCR	CURRENT PROTECTION RELAY			-BGE14, -BGE15	POSITION SWITCHES FOR OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS, ACTUATED BY ELECTRIC MOTOR			
	-BCS1	CURRENT SENSOR LOCATED ON PHASE L1			-BGF	POSITION SWITCHES OF MEDIUM VOLTAGE FUSES			
	-BCS2	CURRENT SENSOR LOCATED ON PHASE L2			-BGI1	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION			
	-BCS3	CURRENT SENSOR LOCATED ON PHASE L3			-BGI2	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN CLOSED POSITION			
	-BCT1	CURRENT TRANSFORMER LOCATED ON PHASE L1			-BGI3	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)			D
	-BCT2	CURRENT TRANSFORMER LOCATED ON PHASE L2			-BGI4	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR			
	-BCT3	CURRENT TRANSFORMER LOCATED ON PHASE L3			-BGI5	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR			
	-BCT4,-BCT7	CURRENT TRANSFORMER LOCATED ON PHASE L1			-BGI6	POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN EMERGENCY MANUAL OPERATION			
	-BCT5,-BCT8	CURRENT TRANSFORMER LOCATED ON PHASE L2			-BGI7	POSITION SWITCH SIGNALLING DICONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN MANUAL OPERATION			
	-BCT6,-BCT9	CURRENT TRANSFORMER LOCATED ON PHASE L3			-BGI8	POSITION SWITCHES OF SWITCH-DISCONNECTOR -QBS			
E	-BCZ	DISTANCE PROTECTION RELAY			-BGI9	POSITION SWITCHES OF OPERATION OF SWITCH-DISCONNECTOR -QBS ACTUATED BY ELECTRIC MOTOR			
	-BEF	FREQUENCY PROTECTION RELAY			-BGI11	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION			E
	-BER	SUPERVISION RELAYS			-BGI12	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN CLOSED POSITION			
	-BES	SYNCHRONIZING RELAY			-BGI13	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)			
	-BET	THERMAL PROTECTION RELAY			-BGI14, -BGI15	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR			
	-BGA1	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT			-BGI21	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION			
	-BGA2	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)			-BGI22	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN CLOSED POSITION			
	-BGA3	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT			-BGI23	POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)			
	-BGA4	POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)			-BGI24, -BGI25	POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR			
	-BGB1...-BGB3	POSITION SWITCHES OF CIRCUIT-BREAKER							
-BGB4	POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS								
F									F
				Date Prep.	15.11.2018	PROJECT TITLE			
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant			
0	First Issued	19.11.2018	YR	Checked	AW				
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A	-BGI31		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION						-CA	CAPACITORS						A
	-BGI32		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION						-CA1	CAPACITOR FOR FRONT VENTILATOR						
	-BGI33		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)						-CA2	CAPACITOR FOR REAR VENTILATOR						
	-BGI41...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION						-EA1	LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.						
	-BGI42		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION						-EA2	LIGHTING LAMP LOCATED IN CABLE COMPARTMENT						
B	-BGI43...		POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION						-EA3	LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE						B
	-BGI51		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION						-EA4	LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT						
	-BGI52		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION						-EB1	HEATER LOCATED IN CABLE COMPARTMENT						
	-BGI53		POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION						-EB2	HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT						
	-BGK		POSITION SWITCH OF KEY LOCK						-EB3	HEATER LOCATED IN MOTOR CABINET						
C	-BGL1		POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1						-EB4	HEATER FOR VENTILATION TEST						C
	-BGS1....-BGS4		POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS						-EB5	HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.						
	-BGS5		PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION						-EB7	HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE						
	-BGT1		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION						-EB8	HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE						
	-BGT2		POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION						-FA1	SURGE ARRESTER LOCATED ON PHASE L1						
D	-BGT3		POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION						-FA2	SURGE ARRESTER LOCATED ON PHASE L2						D
	-BGT4		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION						-FA3	SURGE ARRESTER LOCATED ON PHASE L3						
	-BGT5		POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION						-FCD	FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS						
	-BGT6		POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION						-FCF1	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1						
	-BGT7		POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION						-FCF2	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2						
E	-BGT8		POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION						-FCF3	MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3						E
	-BGT11		PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION						-FCM1	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER						
	-BGT12		PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION						-FCM2...-FCM10	MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS						
	-BGV1		POSITION SWITCH OF FRONT VENTILATOR						-FCM11	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1						
	-BGV2		POSITION SWITCH OF REAR VENTILATOR						-FCM12	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2						
F	-BM		HYGROSTAT						-FCM13	MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3						F
	-BPA4		PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-FCT	THERMAL OVERLOAD RELEASES						
	-BPA5		PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GA	GENERATORS						
	-BPA6		PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ1	FRONT VENTILATOR						
	-BPA7		PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING						-GQ2	REAR VENTILATOR						
-BPS1		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER						-GQ3	VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK							
-BPS2		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2						-KFA1	AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE							
-BPS3		PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3						-KFA2	AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE							
-BR		FLAME DETECTORS, SMOKE DETECTORS						-KFA3....-KFA9	AUXILIARY RELAYS OR CONTACTORS							
-BT		THERMOSTAT						-KFC	CLOSING RELAYS OR CONTACTORS							
-BT1		THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS						-KFI	INTEGRATED CIRCUITS							
-BT2		THERMOSTAT FOR ENERGIZATION OF VENTILATORS						-KFL	LOCKOUT RELAY							
-BT3		THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM						-KFM	MICROPROCESSORS							
-BU		BUCHHOLZ RELAY						-KFN	ANTIPUMPING RELAYS							
-BX1		UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING						-KFO	OPENING RELAYS OR CONTACTORS							
-BX2		ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING						-KFP	PROGRAMMABLE LOGIC CONTROLLERS (PLC)							
-BZ,-BZ4		MULTIFUNCTION SENSORS						-KFR	RECLOSING RELAYS							
-BZ1		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1						-KFS	SYNCHRONIZING DEVICES							
-BZ2		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2						-KFT	AUXILIARY TIME RELAYS, DELAY ELEMENTS							
-BZ3		COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3						-KFZ	CONTROLLERS							
								-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER							
								-MAD	MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD							
								-MAD1	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1							
								-MAD2	MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2							
								-MAD9	MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS							
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A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE	-QBH	MANUAL CIRCUIT-BREAKERS
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2	-QBL	LINKS
B	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS	-QBM	MINIATURE SWITCH-DISCONNECTORS
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING	-QBS	SWITCH-DISCONNECTORS
C	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT	-QBT	TRUCK
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER	-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING
D	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS	-QCE	EARTHING SWITCH
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER	-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)
E	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER	-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER	-QQ	CLUTCHES
F	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS	-RAA	ANTIFERRORESONANCE RESISTOR
	-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER	-RAD	DIODES
G	-PFB	BLUE SIGNAL LAMPS	-RAI	INDUCTORS
	-PFF	FLAG RELAYS	-RAR	RESISTORS
H	-PFG	GREEN SIGNAL LAMPS	-RF	FILTERS
	-PFR	RED SIGNAL LAMPS	-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
I	-PFS	SHORT CIRCUIT INDICATORS	-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
	-PFV	VOLTAGE INDICATORS	-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE
J	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE	-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE	-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD
K	-PFW	WHITE SIGNAL LAMPS	-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1
	-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS	-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2
L	-PFY	YELLOW SIGNAL LAMPS	-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2
	-PGA	AMMETERS	-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS
M	-PGC	COUNTERS	-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4
	-PGF	FREQUENCYMETERS	-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5
N	-PGH	HOURMETERS	-SFA	AMMETRIC SWITCHES
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE	-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS
O	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)	-SFL	LOCKING CONTACTS
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)	-SFO	OPENING PUSH-BUTTONS
P	-PGJ	ACTIVE ENERGY METERS	-SFR	RESET PUSH-BUTTONS
	-PGK	REACTIVE ENERGY METERS	-SFS	SELECTOR SWITCHES
Q	-PGM	MULTIFUNCTION INDICATORS	-SFT	TEST PUSH-BUTTONS
	-PGP	POWER-FACTOR METERS	-SFU	UNLOCKING PUSH-BUTTONS
R	-PGQ	VARMETERS	-SFV	VOLTMETRIC SWITCHES
	-PGS	SYNCHRONOSCOPES	-TA	POWER TRANSFORMERS
S	-PGV	VOLTMETERS	-TB	CONVERTER
	-PGW	WATTMETERS	-TB1	RECTIFIER FOR FIRST OPENING RELEASE
T	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)	-TB2	RECTIFIER FOR SECOND OPENING RELEASE
	-QAB	CIRCUIT-BREAKERS	-TB3	RECTIFIER FOR CLOSING RELEASE
U	-QAC	CONTACTORS (FOR POWER)	-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING
	-QBD	DISCONNECTORS	-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT
V	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)	-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER
W	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)		
	-QBD4	TWO-WAY DISCONNECTOR		
X	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER		

				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	REFERENCE DESIGNATIONS	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW			PT. ABB Sakti Industri	MVD-27 20kV Tank Farm	Drawing Nbr.		Sh.	6
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042			Customer Doc. Nbr.		Cont.	16	

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A	-TFA		ACTIVE POWER TRANSDUCERS		B	-XD11		TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS		C
	-TFC		CURRENT TRANSDUCERS			-XD12		TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING		
	-TFF		FREQUENCY TRANSDUCERS			-XD13		TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION		
	-TFJ		ACTIVE ENERGY TRANSDUCERS			-XD14		TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS		
	-TFK		REACTIVE ENERGY TRANSDUCERS			-XD16		TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS		
	-TFM		MULTIFUNCTION TRANSDUCERS			-XD17		TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS		
	-TFP		POWER-FACTOR TRANSDUCERS			-XD18		TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS		
	-TFQ		REACTIVE POWER TRANSDUCERS			-XD19		TERMINAL BLOCK FOR SPECIAL USE		
	-TFS		SIGNAL CONVERTERS			-XDJ		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD		
	-TFV		VOLTAGE TRANSDUCERS			-XDJ1		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS		
	-WA		BUSBARS			-XDM		SEALABLE TERMINAL BLOCK FOR MEASUREMENT		
	-WBC		POWER CABLES			-XDS		CURRENT SOCKETS		
	-WE		EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.			-XDT		TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB		
	-WGC		CONTROL CABLES			-XDU		CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK		
	-WGS		COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)			-XDU1		TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK		
	-WH		OPTICAL FIBRES			-XDV		TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS		
	-XDA		TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS			-XDV1		CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1		
	-XDA1		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1			-XDV2		CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2		
	-XDA2		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2			-XDV3		CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3		
	-XDA3		CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3			-XDV4		CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR		
-XDA5		CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER		-XDX		SUPPORT TERMINAL BLOCK				
-XDB		CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS		-XE1		EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE				
-XDB1		CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)		-XE2		EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE				
-XDB2		CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER		-XE5		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE				
-XDB10...-XDB89		CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS		-XE6		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE				
-XDB7		CONNECTOR FOR TRUCK POSITION CONTACTS		-XE7		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE				
-XDB91		CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR		-XE8		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE				
-XDB92		CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR		-XE9		EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT				
-XDB93		CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR								
-XDB95		CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET								
-XDC		CUSTOMER TERMINAL BLOCK								
-XDC1		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER								
-XDC2		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR								
-XDC3		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT								
-XDC4		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS								
-XDC5		CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER								
-XDE		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE								
-XDE1		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE								
-XDE2		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2								
-XDE3		TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2								
-XDF		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS								
-XDF1		CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR								
-XDF2		CONNECTOR FOR REAR VENTILATOR CIRCUITS								
-XDF3		CONNECTOR FOR FRONT VENTILATOR CIRCUITS								
-XDH		TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)								
-XDI		TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)								

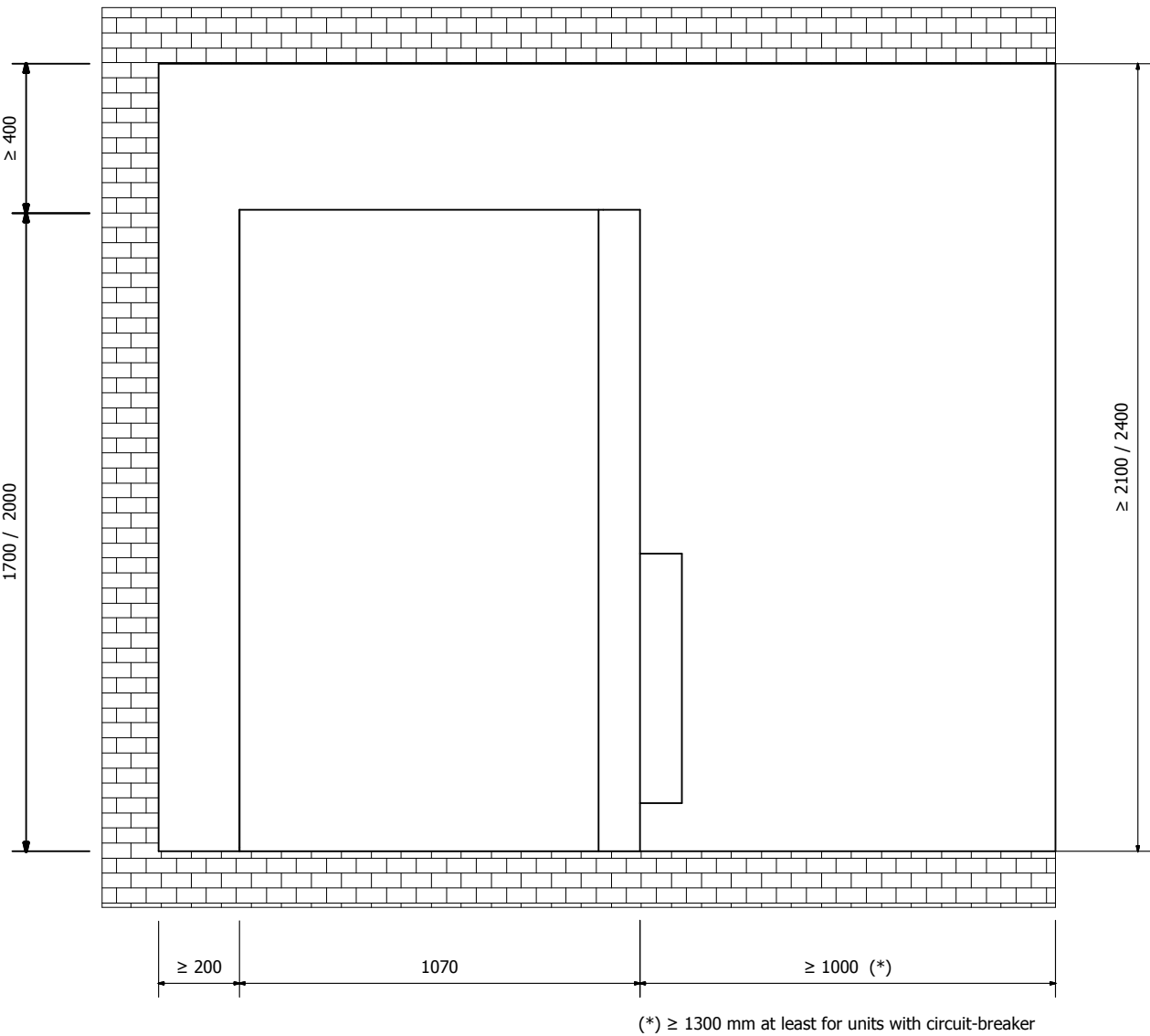
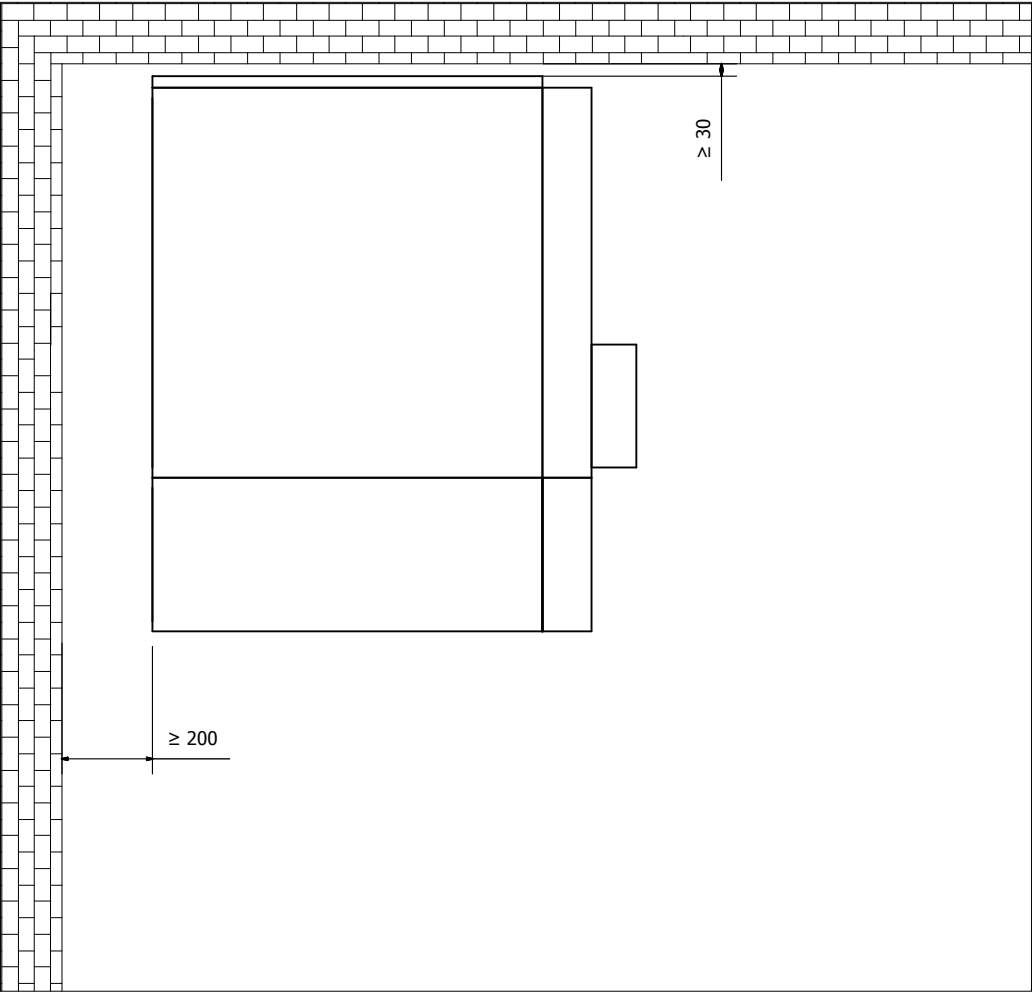
Panel	H01	H02	H03
Panel Name	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming from MVD-25	SBC-W - Outgoing TR 27A Tank Farm&Unloading 2000kVA	DRC - Bus PT
Rated Current	630A	630A	-
Max. Cable Size	2x300mm ²	2x300mm ²	-
Circuit-Diagram	304	306	308

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	 	SINGLE LINE DIAGRAM SLD & GA Drawing MVD-27 20kV Tank Farm	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR				Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW				Drawing Nbr.		Sh. 8
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 16

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RULES FOR INSTALLATION IN THE ROOM

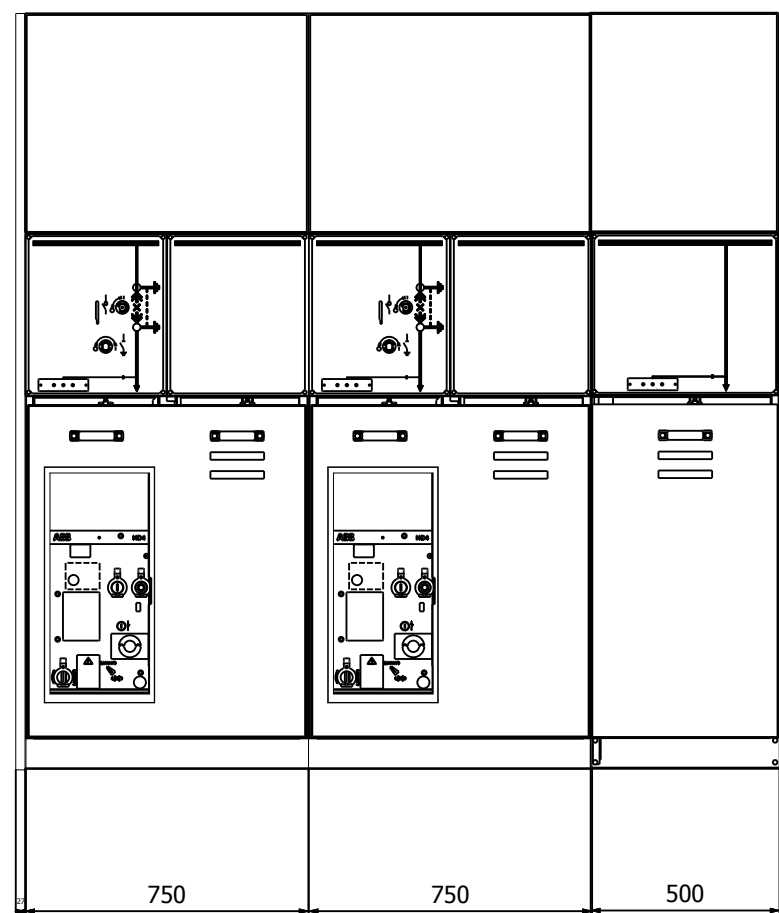
MINIMUM DISTANCES TO BE RESPECTED DURING INSTALLATION



- IAC A-F SOLUTION UP TO 16kA 1sec
- ATTENTION : NO ACCESS TO REAR AND LATERAL OF THE SWG ROOM WHILE SWG IS IN SERVICE

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant			INSTALLATION RULES SLD & GA Drawing MVD-27 20kV Tank Farm	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.			Sh. 9	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		Customer Doc. Nbr.			Cont. 16	

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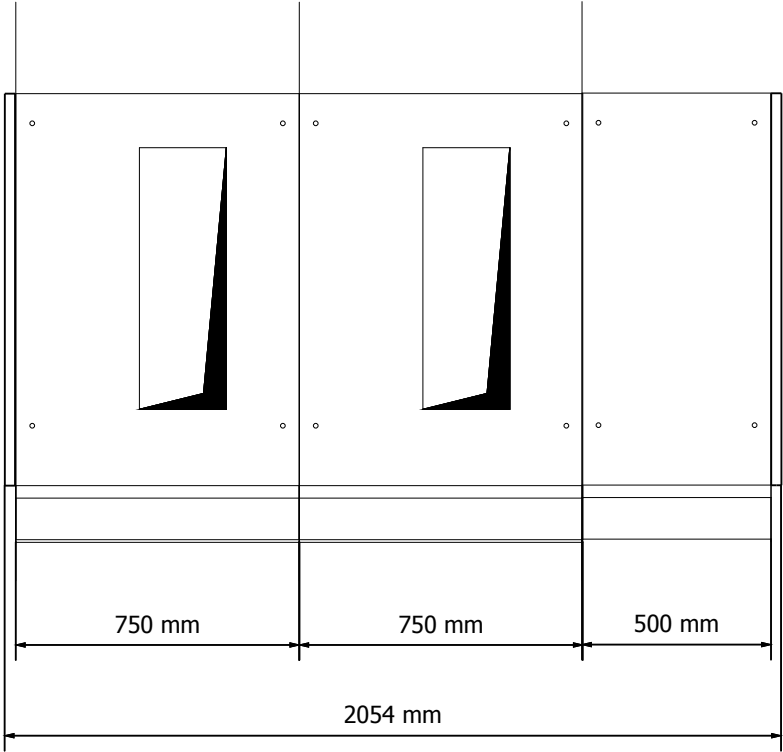


Panel	H01	H02	H03
Panel Type	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming	SBC-W - Outgoing TR 1600kVA	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FRONT VIEW & GA SLD & GA Drawing MVD-27 20kV Tank Farm	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.				Sh. 10	
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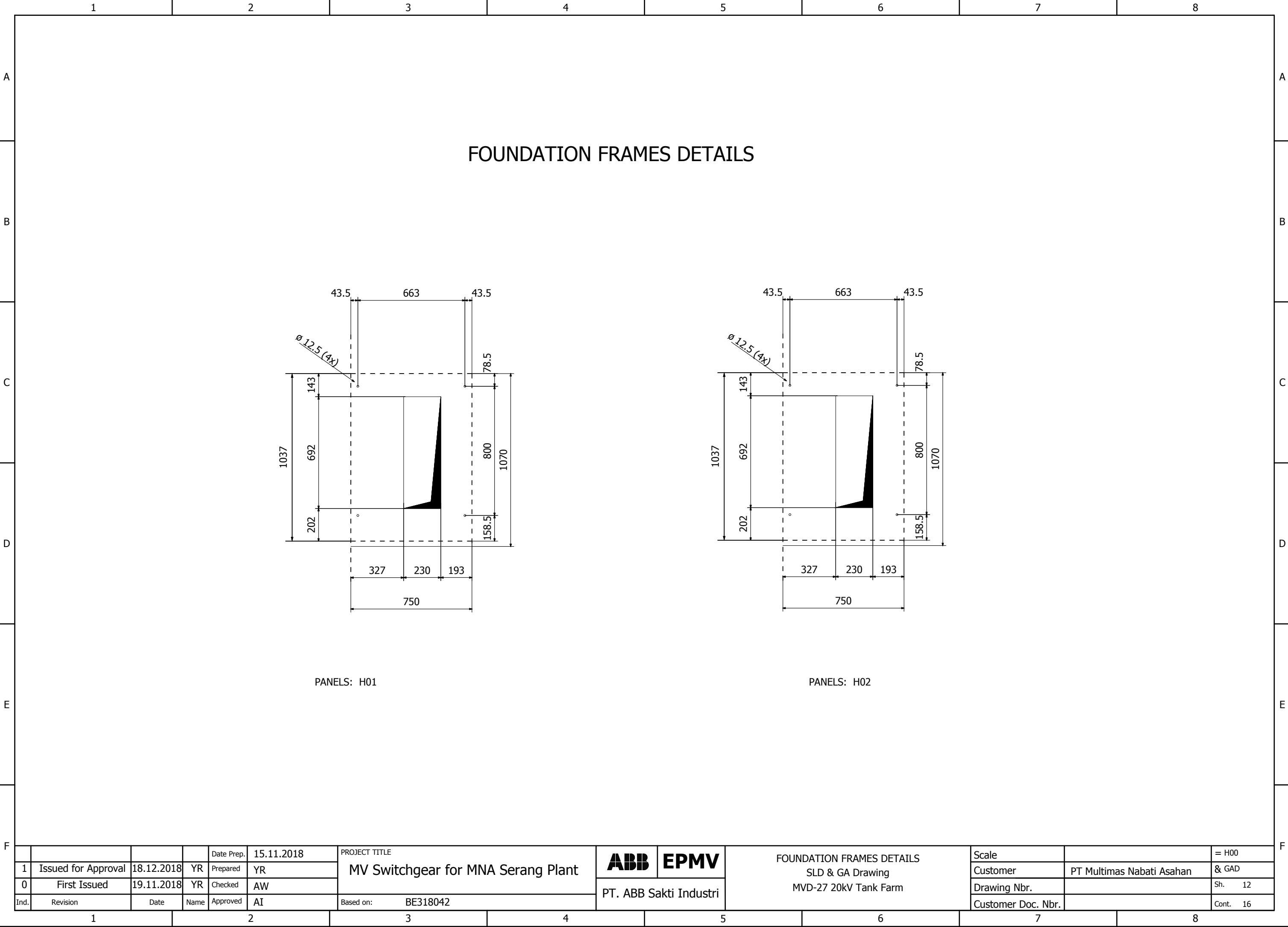
FOUNDATION FRAMES



Panel	H01	H02	H03
Panel Type	SBC-W	SBC-W	DRC
Description	SBC-W - Incoming	SBC-W - Outgoing TR 2000kVA	DRC - Bus PT

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	FOUNDATION FRAMES SLD & GA Drawing MVD-27 20kV Tank Farm	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 11		
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1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 12		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16

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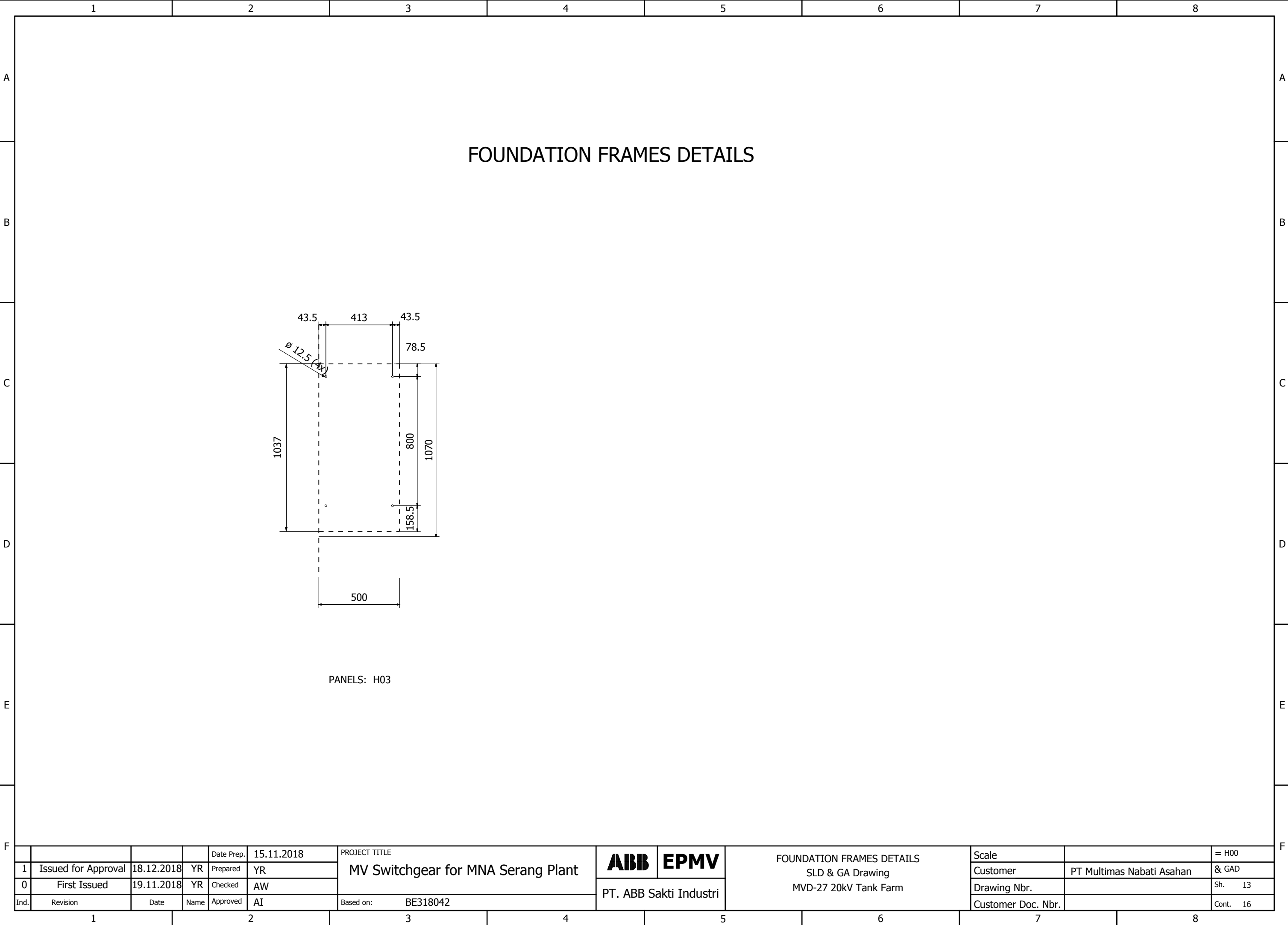
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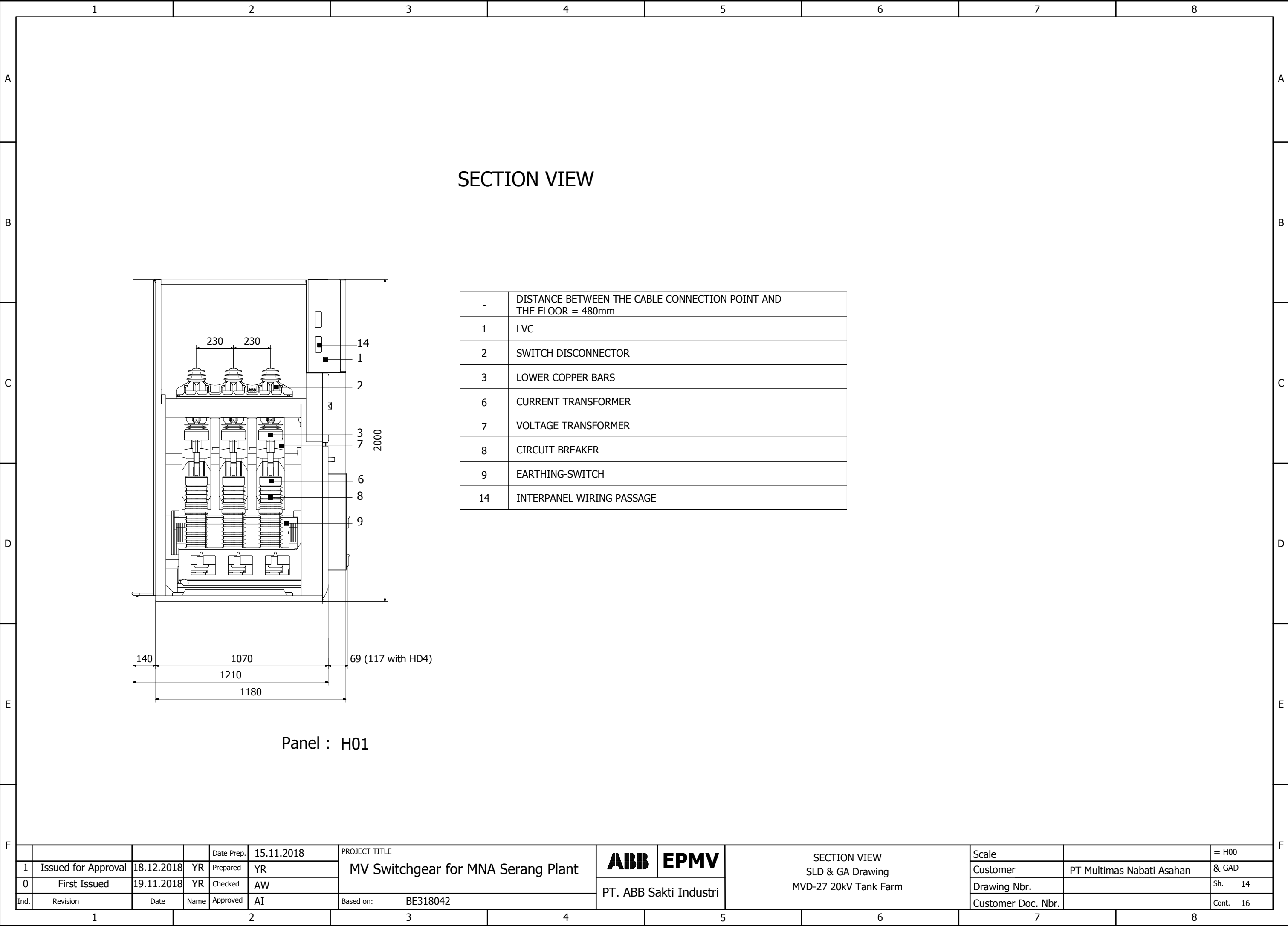
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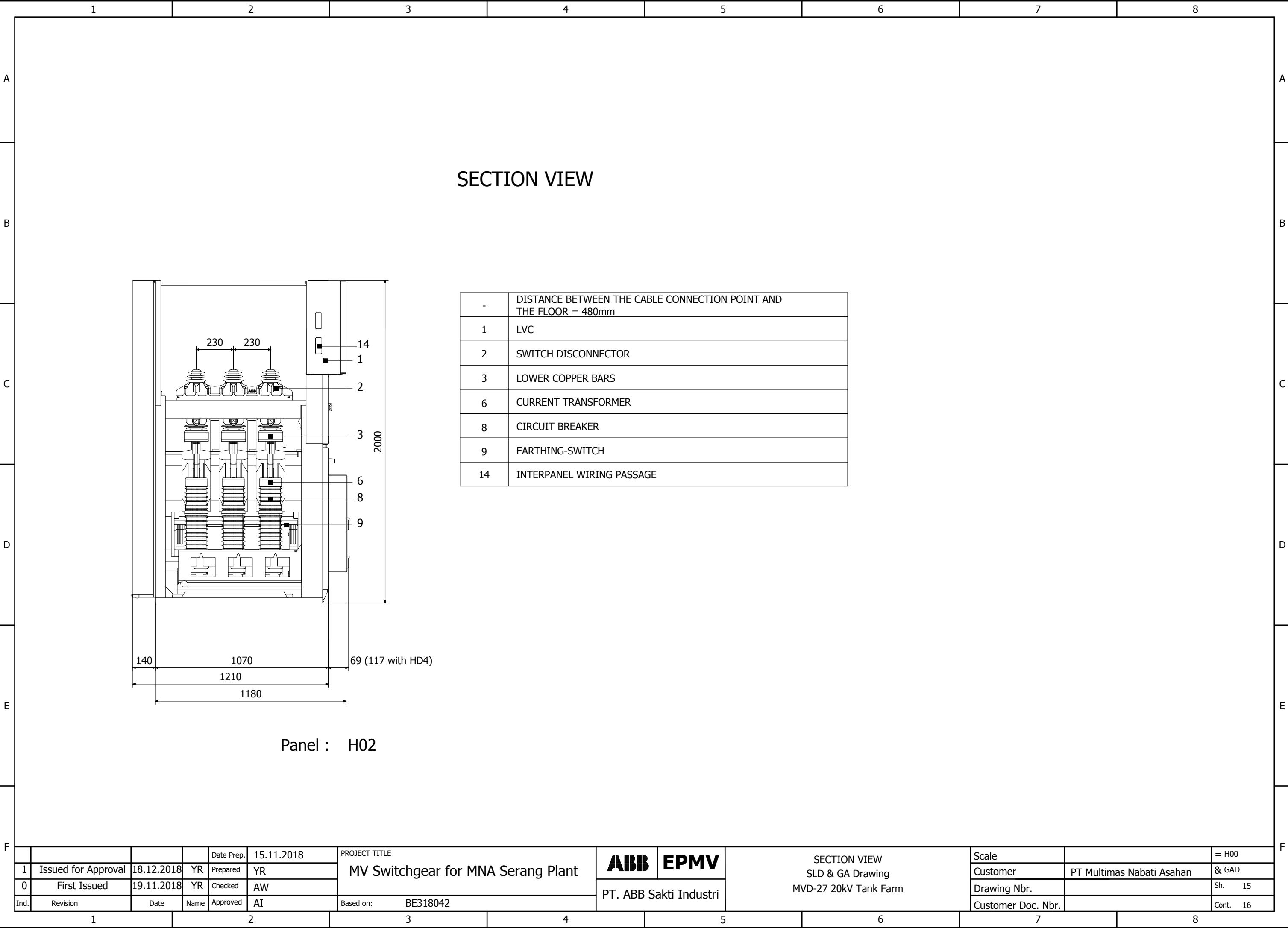
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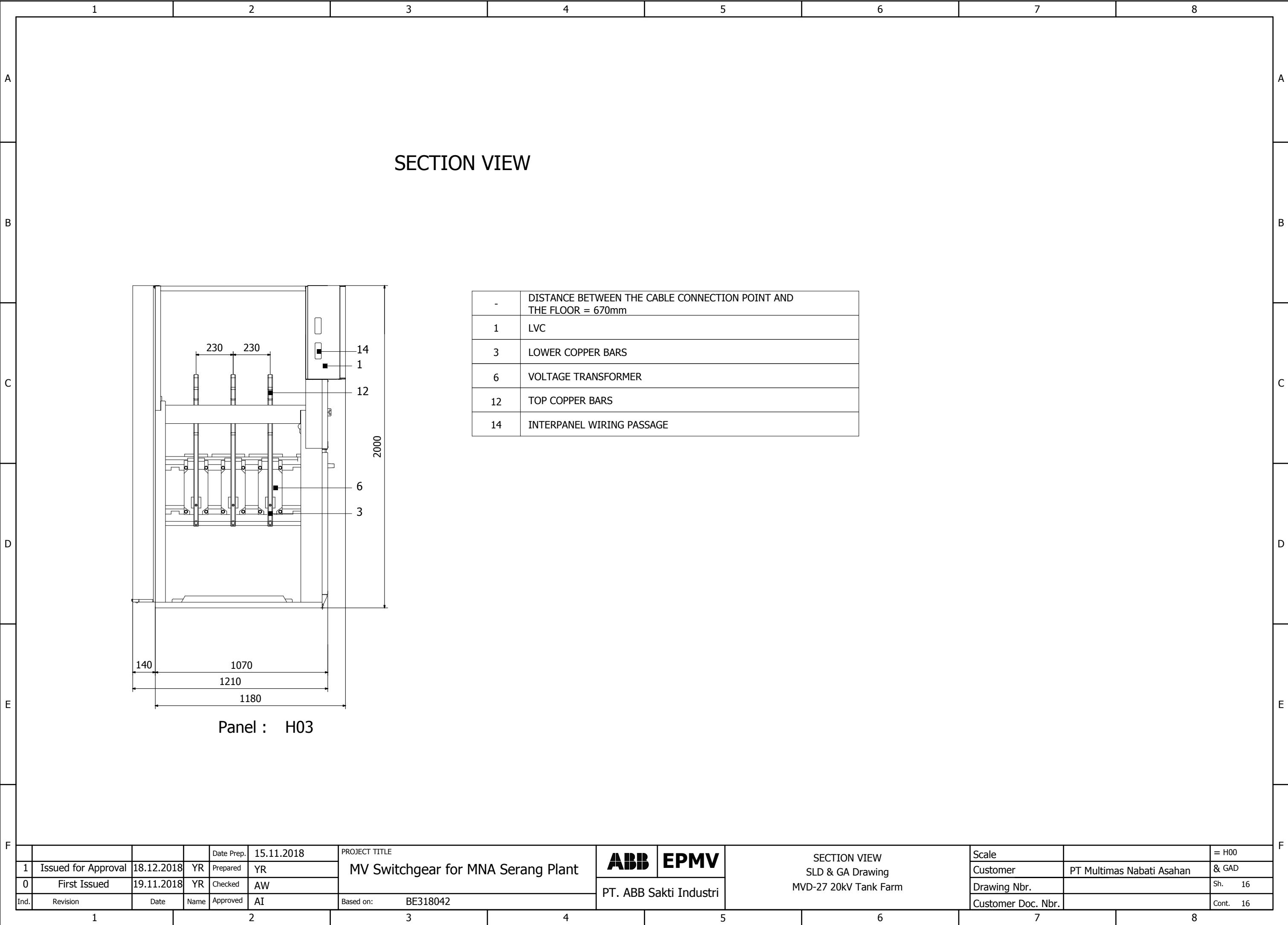
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				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	SECTION VIEW SLD & GA Drawing MVD-27 20kV Tank Farm	Scale		= H00
1	Issued for Approval	18.12.2018	YR	Prepared	YR					Customer	PT Multimas Nabati Asahan	& GAD
0	First Issued	19.11.2018	YR	Checked	AW		Drawing Nbr.			Sh. 16		
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 16